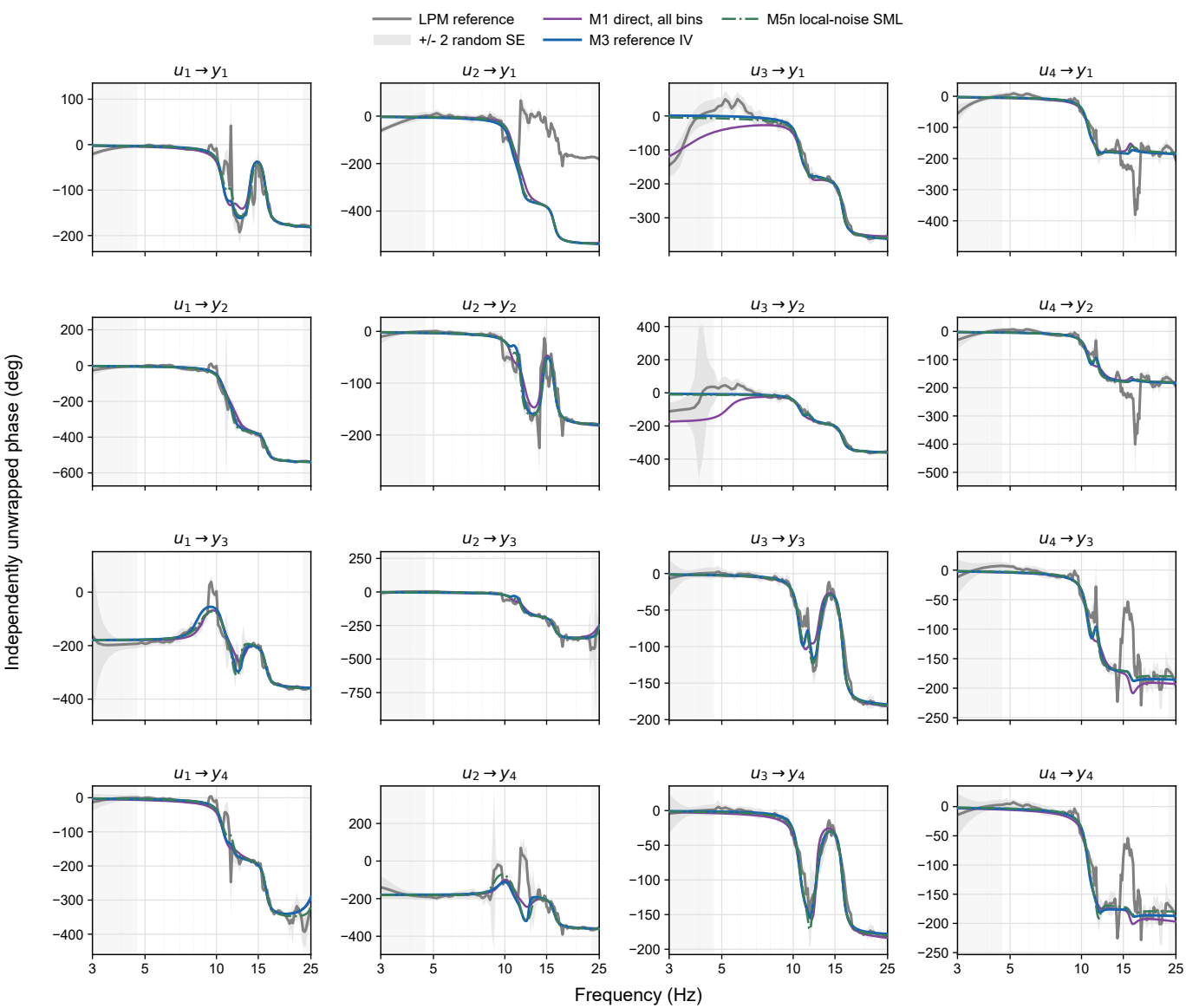

Phase comparison with a nonparametric FRF reference
z = 0 mm | Same-record check: 8-period models / 8-period reference
LPM reference
+/- 2 random SE
M1 direct, all bins
M3 reference IV
M5n local-noise SML
u1 → y1
u2 → y1
u3 → y1
u4 → y1
u1 → y2
u2 → y2
u3 → y2
u4 → y2
u1 → y3
u2 → y3
u3 → y3
u4 → y3
u1 → y4
u2 → y4
u3 → y4
u4 → y4
Independently unwrapped phase (deg)
Frequency (Hz)
LPM is an estimate, not ground truth; smoothing and inversion bias are not included in random SE.
Each curve unwrapped separately; only a constant multiple of 360 deg aligns its first point.